Unit III

3 Communication Protocols and IoT Challenges

3.1 : IEEE 802.11

Q.1 What is WLAN ? Explain 802.11a, 802.11b and 802.11g standard

Ans. : • Wireless Local Area Network (WLAN) has data transfer speeds ranging from 1 to 54 Mbps. WLAN signal can be broadcast to cover an area ranging in size from a small office to a large campus. Most commonly, a WLAN access point provides access within a radius of 65 to 300 feet.

- IEEE 802.11 standard for WLANs.
- The 802.11 specifications were developed specifically for Wireless Local Area Networks (WLANs) by the IEEE and include four subsets of Ethernet-based protocol standards : 802.11, 802.11a, 802.11b, and 802.11g.
- 802.11 operated in the 2.4 GHz range and was the original specification of the 802.11 IEEE standards. This specification delivered 1 to 2 Mbps using a technology known as Phase-Shift Keying (PSK) modulation.
- 802.11a operates in the 5 - 6 GHz range with data rates commonly in the 6 Mbps, 12 Mbps, or 24 Mbps range. Because 802.11a uses the Orthogonal Frequency Division Multiplexing (OFDM) standard, data transfer rates can be as high as 54 Mbps.
- The 802.11b standard operates in the 2.4 GHz range with up to 11 Mbps data rates and is backward compatible with the 802.11 standard. 802.11b uses a technology known as **Complementary Code Keying (CCK)** modulation.

- 802.11g is the most recent IEEE 802.11 draft standard and operates in the 2.4 GHz range with data rates as high as 54 Mbps over a limited distance.

Q.2 Explain hidden and exposed terminal problem.

Ans. :

Hidden Terminal Problem

- Station B can hear both A and C, but A and C can't hear each other.
- It happens when station C attempts to transmit while A is transmitting to B.
- Station "A" is hidden from station C.

Fig. Q.2.1 Hidden station problem

Exposed Terminal :

- Happens when station B is transmitting to A when C attempts to transmit.
- Assuming no interference effect, station C should defer transmitting only if it want to transmit to B.
- Carrier sense provides information about potential collision at the sender, but not at the receiver.

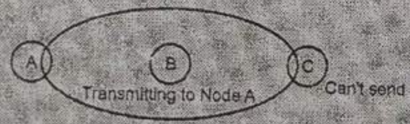


Fig. Q.2.2 Exposed station problem

- To avoid collision, sender senses the carrier before transmission. But collision occurs at the receiver not transmitter.

Q.3 Explain wireless LAN protocols.

Ans. : 1. Multiple Access with Collision Avoidance (MACA)

- MACA protocol solved hidden, exposed terminal. Send Ready-to-Send **(RTS)** and Clear-To-Send **(CTS)** first. RTS, CTS helps determine who else is in range or busy (Collision avoidance).

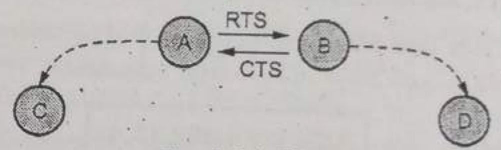


Fig. Q.3.1 MACA

- MACA uses exchange of two short messages : Request To Send (RTS), and Clear To Send (CTS).
- They are fixed size. When A wishes to transmit to B, it sends an RTS message. RTS message contains duration of proposed transmission.
- If B knows that the channel is free, it responds with a CTS message. (CTS also contains duration of proposed communication).
- Any station that hears the RTS message, defers all communication for some time until the associated CTS message has been finished.
- A CTS message defers communication for the duration of the time indicated in the CTS message.
- When A is transmitting data, C can go ahead and access the channel. Node B's CTS message may not be heard by A. Station B found that the channel was already busy. RTS packet might collide.
- If A does not receive a CTS, it times-out and schedules the packet for retransmission. MACA uses the binary exponential back-off algorithm to select the retransmission time. B's CTS message collides at C.
- This would cause C to be unaware of the pending communication between nodes A and B.

2. Multiple Access with Collision Avoidance for Wireless LANs (MACAW)

- Receiver responds with Acknowledgment (ACK) after data reception. Other nodes in receiver's range learn that channel is available. Nodes hearing RTS, but not CTS do not know if transmission will occur.
- MACAW uses Data Sending (DS) packet, sent by sender after receiving CTS to inform such nodes of successful handshake.

3.2 : IEEE 802.15.4

Q.4 Write a short note on IEEE 802.15.4 access technology.

[SPPU, May-19, End Sem, Marks 5]

Ans. : • 802.15.4 is an IEEE standard for PHY and MAC layers upon which further networking and communication protocols are built. Networking standards built upon 802.15.4 include Thread and Zigbee.

- The IEEE 802.15.4 protocol is designed for enabling communication between compact and inexpensive low power embedded devices that need a long battery life.
- IEEE 802.15.4 is simple packet data protocol for lightweight wireless networks. Channel access is via Carrier Sense Multiple Access with collision avoidance and optional time slotting. It provides multi-level security.
- **Characteristic of 802.15.4 :**

1. Data rates of 250 kbps, 20 kbps and 40 kbps.
2. Star or peer-to-peer operation.
3. Support for low latency devices.
4. CSMA-CA channel access.
5. Dynamic device addressing.
6. Fully hand-shaked protocol for transfer reliability.
7. Low power consumption.

8. 16 channels in the 2.4 GHz ISM band, 10 channels in the 915 MHz ISM band and one channel in the European 868 MHz band.

- The IEEE 802.15.4 supports two classes of devices : Fully Functional Devices (FFD), which have full network functionalities and the Reduced Functional Devices (RFD), which possess limited functionalities.
- All Personal Area Networks (PAN) consists of at least one FFD which acts as the PAN coordinator which is responsible for maintaining the PAN. RFDs are responsible for directly obtaining data from the environment and sending them to a PAN coordinator.
- IEEE 802.15.4 defines four types of frames :

a) Beacon frames, b) MAC command frames,
c) Acknowledgement frames, d) Data frames.

Q.5 What are the functions of physical and MAC layer of 802.15.4 ?

Ans. : • Functions of physical layer :

1. Activation and deactivation of the radio transceiver.
2. Energy detection within the current channel.
3. Link quality indication for received packets.
4. Clear channel assessment for CSMA-CA.
5. Channel frequency selection.
6. Data transmission and reception.

- **The IEEE 802.15.4 MAC layer is responsible for :**

1. Joining and leaving the PAN;
2. Carrier Sense Multiple Access with Collision Avoidance (CSMA-CA) for channel access;
3. Guaranteed Time Slot (GTS) transmissions;
4. Establishing a reliable link between two peer MAC entities;
5. Beacon transmissions for a coordinator;
6. Synchronization to the beacons.

Dec 06, 2023, 19:12

Q.6 What is beacon enable mode ?

Ans. : • All IEEE 802.15.4-based networks use beacons from a co-ordinator when joining devices to the network.

- The co-ordinator sends out a periodic train of beacon signals containing information that allows network nodes to synchronise their communications. A beacon also contains information on the data pending for the different nodes of the network.
- Normally, two successive beacons mark the beginning and end of a superframe. A superframe contains 16 timeslots that can be used by nodes to communicate over the network. Fig. Q.6.1 shows superframe.

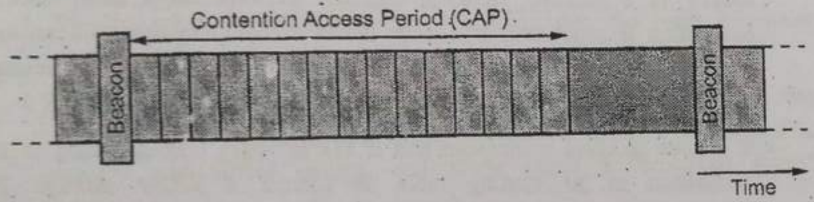


Fig. Q.6.1 Superframe

- The total time interval of these timeslots is called the **Contention Access Period (CAP)**, during which nodes can attempt to communicate using slotted CSMA/CA.

3.3 : Bluetooth and ZigBee

Q.7 Write short note on Bluetooth.

Ans. : • Bluetooth is a wireless technology that enables a wireless device to communicate in the 2.4 GHz Industrial Scientific and Medical (ISM band).

- Bluetooth is a short range, low cost and low power wireless communication standard.
- Bluetooth supports point-to-point and point-to-multipoint connections.
- Bluetooth operates in the 2.4 GHz Industrial, Scientific and Medical (ISM) band at a maximum data rate of 720 kbps. It uses frequency hopping spread spectrum that divides the frequency band yielding

79 channels. The modulation used is a Gaussian-shaped binary Frequency Shift Keying (GFSK) modulation scheme.

- Bluetooth uses TDD (Time Division Duplex) to perform full-duplex communication. The master defines slots of time and allocates one slot per slave.
- Bluetooth is organized in master-slave mode. The topology supports 1 master, 7 slaves and up to 255 slaves in parked mode. The master defines the clock and time slots for all nodes.
- A master can request to change his role with a slave, at which point the slave becomes the master and the master becomes the slave. For example : A head phone starts the communication as a master, but it prefers to be the slave.
- Two basic topologies are possible in Bluetooth : Piconet and Scatternet Topology.
- Bluetooth devices have three modes to save power energy in transmission inactive state : Sniff mode, hold mode and parked mode.

Q.8 What is ZigBee ? List its features.

Ans. : • ZigBee is an open, global, packet-based protocol designed to provide an easy-to-use architecture for secure, reliable, low power wireless networks. Flow or process control equipment can be place anywhere and still communicate with the rest of the system. It can also be moved, since the network doesn't care about the physical location of a sensor, pump or valve.

Features

1. **Stochastic addressing :** A device is assigned a random address and announced. Mechanism for address conflict resolution. Parents node don't need to maintain assigned address table.
2. **Link management :** Each node maintains quality of links to neighbors. Link quality is used as link cost in routing.
3. **Frequency agility :** Nodes experience interference report to channel manager, which then selects another channel.

4. **Asymmetric link :** Each node has different transmit power and sensitivity. Paths may be asymmetric.
5. **Power management :** Routers and co-ordinators use main power. End devices use batteries.

Q.9 Explain the functions of ZigBee coordinator.

Ans. : • Function of ZigBee coordinator :

1. It forms a network.
2. It establishes the 802.15.4 channel on which the network will operate.
3. It establishes the extended and short PAN ID for the network.
4. It decides on the stack profile to use (compile or run-time option).
5. It acts as the trust center for secure applications and networks.
6. It acts as the arbiter for End-Device-Bind (a commissioning option).
7. It acts as a router for mesh routing.
8. It acts as the top of the tree, if tree routing enabled.

Q.10 Explain the following with ZigBee network ?
1. Forming a Network 2. Joining a Network
3. Network Communication

Ans. : 1. Forming a Network

- The coordinator initiates network formation.
- After forming the network, the coordinator can function as a router and can accept requests from other devices wishing to join the network.
- Depending on the stack and application profile used, the coordinator might also perform additional duties after network formation.

2. Joining a Network

- A device finds a network by scanning channels.
- When a device finds a network with the correct stack profile that is open to joining, it can request to join that network.
- A device sends its join request to one of the network's router nodes and the device receiving the request can then use the callback to accept the request or deny it.

3. Network Communication

- All nodes that communicate on a network transmit and receive on the same channel, or frequency.
- ZigBee uses a Personal Area Network Identifier (PAN ID) to identify a network this provides a way for two networks to exist on the same channel while still maintaining separate traffic flow.

Q.11 Write short note on ZigBee-AODV.

Ans. : • Another routing protocol in ZigBee is a route discovery protocol similar to Ad-hoc On-demand Distance Vector Routing.

- AODV is a reactive protocol, i.e., the network stays silent until a connection is requested. If a node wants to communicate with another one, it broadcasts a request to its neighbors who re-route the message and safeguard the node from which they received the message.
- If a node receives a message and it has an entry corresponding to the destination in its routing table, it returns a RREP through the reverse-path to the requesting node. So, the source sends its data through this path to the destination with the minimum number of hops.
- In ZAODV, when a node desires to send data to another node without having the related routing information, it buffers the data and then initiates a route discovery process to find the optimal path for the destination.

  1. **Routing Table (RT)**
  2. **Route Discovery Table (RDT)**
  3. **Route Request Command Frame (RREQ)**
  4. **Route Reply Command Frame (RREP)**

- The ZigBee routing algorithm uses a path cost metric for route comparison during route discovery and maintenance.

RREQ

1. The source node broadcasts a route request (RREQ) packet.
2. If a intermediate router receives an new RREQ, it add RDT entry and rebroadcast the RREQ.

Dec 06, 2023, 19:12

3. If a intermediate router receives an old RREQ, it shall compare the Forward Cost in the RREQ to the one stored in the RDT entry.
4. If the Cost is less, the router shall update the RDT entry and **rebroadcast the RREQ again.**
5. Using Source Address and Route Request ID, the RREQ can be uniquely identified.

RREP

1. The destination node receives the RREQ, it responds by unicasting a route reply (RREP) packet to its neighbor from which it received the RREQ.
2. The destination will choose the routing path with the lowest cost and then send a route reply.
3. Finally, when the RREP reaches the originator, the route discovery process is finished.

- ZAODV can effectively reduce the transmission delay. However, the procedure of flooding RREQ wastes significant amount of bandwidth, memory and device energy, and might cause a "broadcast storm" problem.

Q.12 What is use of APS layer in ZigBee ?

Ans. : • Application Support Sublayer (APS) is responsible for :

1. Binding tables
2. Message forwarding between bound devices
3. Group address definition and management
4. Address mapping from 64-bit extended addresses to 16-bit NWK addresses
5. Fragmentation and reassembly of packets
6. Reliable data transport.

- The key to interfacing devices at the need/service level is the concept of binding. Binding tables are kept by the coordinator and all routers in the network. The binding table maps a source address and source endpoint to one or more destination addresses and endpoints. The cluster ID for a bound set of devices will be the same.

- APS layer is responsible for providing a data service to the application and ZigBee device profiles. It also provides a management service to maintain binding links and the storage of the binding table itself.
- **ZigBee's security services include methods for key establishment and** transport, device management, and frame protection. The ZigBee specification defines security for the MAC, network and APS layers. Security for applications is typically provided through Application Profiles.
- The APS layer allows frame security to be based on link keys or the network key. Encryption can be applied at three different levels, the MAC layer, Network layer and the Application Support (APS) layer. A ZigBee frame will contain fields from all of these layers encapsulated within one another, and each later can encrypt its data payload.
- The APS layer is also responsible for providing applications and the ZDO with key establishment, key transport, and device management services. Fig. Q.12.1 shows Zigbee frame with security on the APS level.

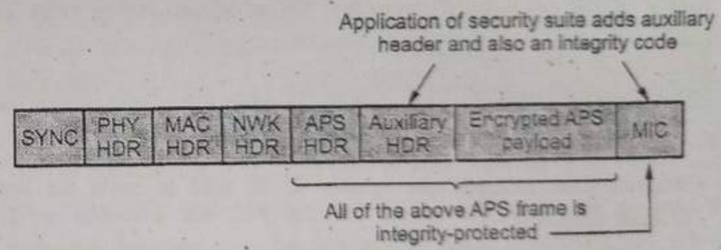


Fig. Q.12.1 Zigbee frame with security on the APS level

- The APS sublayer is key establishment services provide the mechanism by which a ZigBee device may derive a shared secret key, with another ZigBee device. Key establishment involves two entities, an initiator device and a responder device, and is prefaced by a trust-provisioning step.

Dec 06, 2023, 19:13

3.4 : IP Based Protocol

Q.13 Draw and explain in detail IPv4 header format.

Ans. : • The IP datagram consists of a header followed by a number of bytes of data. Fig. Q.13.1 shows header format of IPv4. All IPv4 packets use the 20 bytes header.

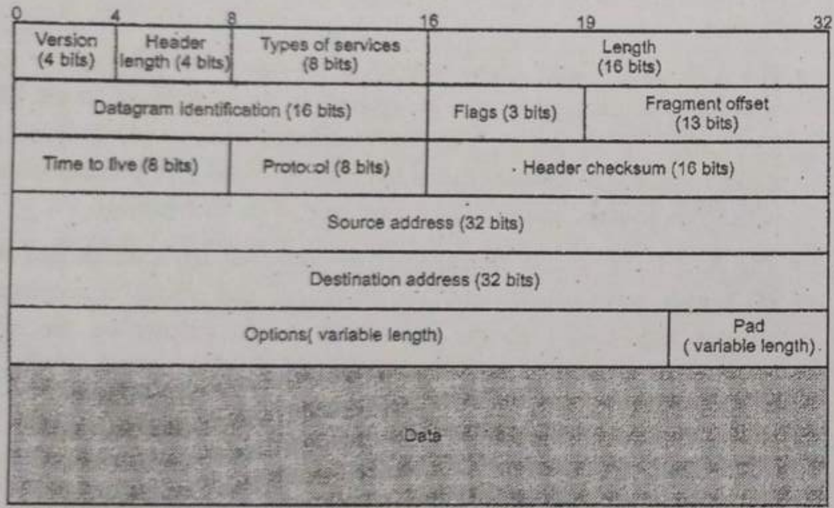


Fig. Q.13.1 : IPv4 header format

- **Version field** indicates the version of IP used to build the header. Current version is 4.
- **Header Length** indicates the length of the IP header in 32 bits words. This field allows IPv4 to use options if required, but as it is encoded as a 4 bits field, the IPv4 header cannot be longer than 64 bytes.
- **Type of service :** The service type is an indication of the quality of service requested for this IP datagram.
- **Protocol field** indicates the transport layer protocol that must process the packet's payload at the destination. Common values for this field are 6 for TCP and 17 for UDP.

- **Length field** indicates the total length of the entire IPv4 packet (header and payload) in bytes. This implies that an IPv4 packet cannot be longer than 65535 bytes.
- **Fragment offset** is used to reassemble the full datagram. The value in this field contains the number of 64-bit segments (header bytes are not counted) contained in earlier fragments. If this is the first (or only) fragment, this field contains a value of zero.
- **Flags (3 bits) :** Only two of the bits are currently defined : MF (More Fragments) and DF (Don't Fragment).
- **More Fragments (MF) flag** is a single bit in the Flag field used with the Fragment Offset for the fragmentation and reconstruction of packets. The More Fragments flag bit is set, it means that it is not the last fragment of a packet. When a receiving host sees a packet arrive with the MF = 1, it examines the Fragment Offset to see where this fragment is to be placed in the reconstructed packet. When a receiving host receives a frame with the MF = 0 and a non-zero value in the Fragment offset, it places that fragment as the last part of the reconstructed packet.
- **Don't Fragment (DF) flag** is a single bit in the Flag field that indicates that fragmentation of the packet is not allowed. If the Don't Fragment flag bit is set, then fragmentation of this packet is NOT permitted. If a router needs to fragment a packet to allow it to be passed downward to the Data Link layer but the DF bit is set to 1, then the router will discard this packet.
- **Time-To-Live (TTL)** is an 8-bit binary value that indicates the remaining "life" of the packet. The TTL value is decreased by at least one each time the packet is processed by a router (that is, each hop). When the value becomes zero, the router discards or drops the packet and it is removed from the network data flow.
- **Source address** field that contains the IPv4 address of the source host.
- **Destination address** field that contains the IPv4 address of the destination host.

Dec 06, 2023, 19:13

- **Checksum** that protects only the IPv4 header against transmission errors. Checksum is for the information contained in the header. If the header checksum does not match the contents, the datagram is discarded.
- **Options (variable) :** Encodes the options requested by the sending user.
- **Padding (variable) :** Used to ensure that the datagram header is a multiple of 32 bits.
- **Data (variable) :** The data field must be an integer multiple of 8 bits. The maximum length of the datagram (data field plus header) is 65,535 octets.

Q.14 What do you mean private and public address ?

Ans. : • IPv4 addresses are further classified as either public or private. Public IP addresses are ones that are exposed to the Internet. Any other computers on the Internet can potentially communicate with them.

- Private IP addresses are hidden from the Internet and any other networks. They are usually behind an IP proxy or firewall device.
- Private addresses are as follows :

Class	Starting address	End of range
Class A	10.0.0.0	10.255.255.255
Class B	172.16.0.0	172.31.255.255
Class C	192.168.0.0	192.168.255.255

Q.15 Compare IPv4 and IPv6.

Ans. :

Sr. No.	IPv4	IPv6
1.	Header size is 32 bits.	Header size is 128 bits.
2.	It cannot support autoconfiguration.	Supports autoconfiguration.
3.	Cannot support real time application.	Supports real time application.
4.	No security at network layer.	Provides security at network layer.
5.	Throughput and delay is more.	Throughput and delay is less.

Q.16 Explain IPv6 address space and transmission modes.

[SPPU : May-19, End Sem, Marks 8]

Ans. : • IPv6 allows three types of addresses : Unicast, Anycast and Multicast.

- **Unicast :** An identifier for a single interface. A packet sent to a unicast address is delivered to the interface identified by that address.
- **Anycast :** An identifier for a set of interfaces. A packet sent to an anycast address is delivered to one of the interfaces identified by the address.
- **Multicast :** An identifier for a set of interfaces. A packet sent to a multicast address is delivered to all interfaces identified by that address.
- The first field of any IPv6 address is the variable - length format prefix, which identifies various categories of address.
- IPv6 is multicasting, which provides efficient network operations.
- IPv6 addresses are 128 bits in length. Addresses are assigned to individual interface on nodes, not to the node themselves.
- IPv6 addresses are assigned to interfaces, rather than to nodes, in recognition that a node can have more than one interface.
- A single interface may have multiple unique unicast addresses. The first field of any IPv6 address is the variable length format prefix, which identifies various categories of addresses.

Dec 06, 2023, 19:13

- A new notation has been devised for writing 16-byte addresses. They are written as eight groups of four hexadecimal digits with colons between the groups, like this

  8000 : 0000 : 0000 : 0000 : 0123 : 4567 : 89AB : CDEF

Q.17 Describe IPv6 packet format and compare IPv4 & IPv6 headers.
[SPPU, May-19, End Sem, Marks 8]

Ans. : • Fig. Q.17.1 the IPv6 datagram header format. Each packet is composed of a mandatory base header followed by the payload.

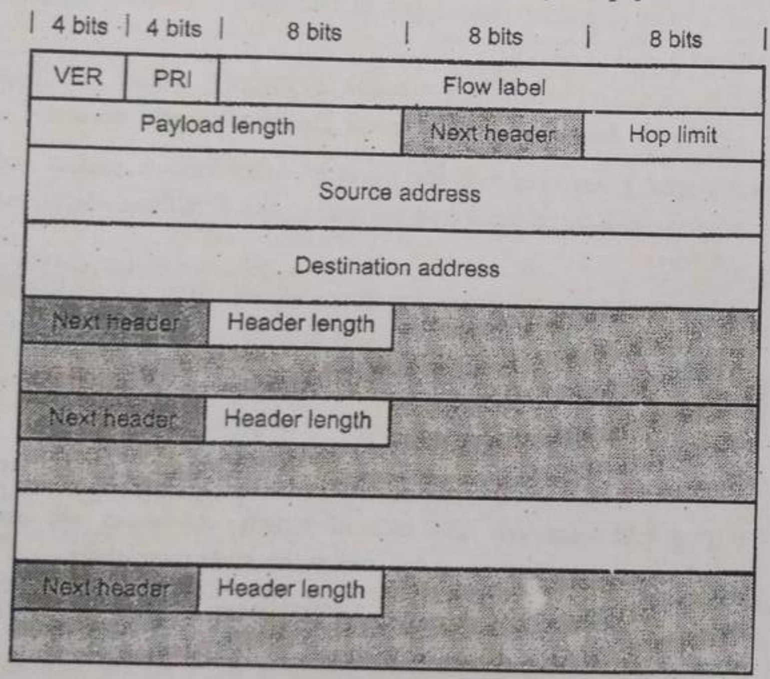


Fig. Q.17.1 : IPv6 header

- The payload consists of two parts : Optional and data.

1. **Versions :** This 4 bits field defines the version number of the IP. The value is 6 for IPv6.
2. **Priority :** The 4 bits priority field defines the priority of the packet with respect to traffic congestion.

3. **Flow label :** It is 24 bits field that is designed to provide special handling for a particular flow of data.
4. **Payload length :** The 16 bits payload length field defines the length of the IP datagram excluding the base header.
5. **Next header :** It is an 8 bits field defining the header that follows the base header in the datagram.
6. **Hop limit :** This 8 bits hop limit field serves the same purpose as the TTL field in IPv4.
7. **Source address :** The source address field is a 128 bits internet address that identifies the original.
8. **Destination address :** It is 128 bits Internet address that usually identifies the final destination of the datagram.

Q.18 Compare IPv4 and IPv6 headers.

Ans. :

IPv4 headers	IPv6 headers
IPv4 has a header of 20-60 bytes.	IPv6 has header of 40 bytes fixed.
The source and destination IPv4 addresses are 32 bit binary numbers.	Source and destination IPv6 addresses are 128 bit binary numbers.
IPv4 header includes space for IPv4 options.	In IPv6 header, similar feature known as extension header.
In the IPv4 header, there is a Checksum field that is used to verify and discard corrupted packets.	In IPv6, here is no checksum field.

Q.19 Write a short note on Goals and header stacks of 6LOWPAN.
[SPPU : May-19, End Sem, Marks 8]

Ans. : Goals :

- Define adaptation (frag/reassembly) layer to match IPv6 MTU requirements.
- Specify methods to do IPv6 stateless address auto configuration.

Dec 06, 2023, 19:13

- Specify/use header compression schemes.
- Specify implementation considerations and best methods of an IPv6 stack.
- Methods for meshing on LoWPAN below IP.

Header Stacks :

- 6LoWPAN uses stacked headers and analogous to IPv6, extension headers. 6LoWPAN headers define the capability of each sub-header.
- Three sub-headers are defined : Mesh addressing, fragmentation and header compression.
- Fig. Q.19.1 shows 6LoWPAN stacked headers.

802.15.4 Header	IPv6 Header compression	IPv6 Payload

802.15.4 Header	Fragment header	IPv6 Header compression	IPv6 Payload

802.15.4 Header	Mesh addressing header	Fragment header	IPv6 Header compression	IPv6 Payload

Fig. Q.19.1 6LoWLAN header stack

- Mesh addressing supports layer-two forwarding and fragmentation supports the IPv6 minimum MTU requirement. 6LoWPAN identifies all header formats using a header type field placed at the beginning of each header.
- A 6LoWPAN Not-A-LoWPAN (NALP) type enables it to coexist with other protocols that operate directly on the link. The header stack is simple to parse and allows elision of headers when unneeded.
- The fragmentation header is elided for small datagrams, indicating that a single frame carries the entire payload. Similarly, the mesh header is elided when 6LoWPAN frames are delivered over a single radio hop, so the path source and destination are identical to those in the link-layer header.

Q.20 What is 6LOWPAN? Explain header compression of 6LoWPAN.

Ans. : • 6LoWPAN provides a means of carrying packet data in the form of IPv6 over IEEE 802.15.4 and other networks. It provides end-to-end IPv6 and as such it is able to provide direct connectivity to a huge variety of networks including direct connectivity to the Internet.

- In order to send packet data, IPv6 over 6LowPAN, it is necessary to have a method of converting the packet data into a format that can be handled by the IEEE 802.15.4 lower layer system.
- IPv6 requires the Maximum Transmission Unit (MTU) to be at least 1280 bytes in length. This is considerably longer than the IEEE 802.15.4's standard packet size of 127 octets which was set to keep transmissions short and thereby reduce power consumption.
- To overcome the address resolution issue, IPv6 nodes are given 128-bit addresses in a hierarchical manner. The IEEE 802.15.4 devices may use either of IEEE 64-bit extended addresses or 16-bit addresses that are unique within a PAN after devices have associated. There is also a PAN-ID for a group of physically co-located IEEE 802.15.4 devices.

Header compression

- 6LoWPAN uses header compression mechanisms to reduce the packet overhead when sending data over 802.15.4 links. It effectively reduces the size of the IPv6 header and transport headers sent over-the-air by relying on a set of assumptions and information present in the link layer.
- When communicating an IPv6 packet over IEEE 802.15.4 or a 6LP technology, **a typical 40-byte IPv6 header would consume a significant** fraction of frame payload, as well as precious energy and bandwidth resources.
- 6LoWPAN header compression was defined to produce a lightweight encoding for the IPv6 header over IEEE 802.15.4. 6LoWPAN IC leverages stateless and stateful techniques.

Dec 06, 2023, 19:13

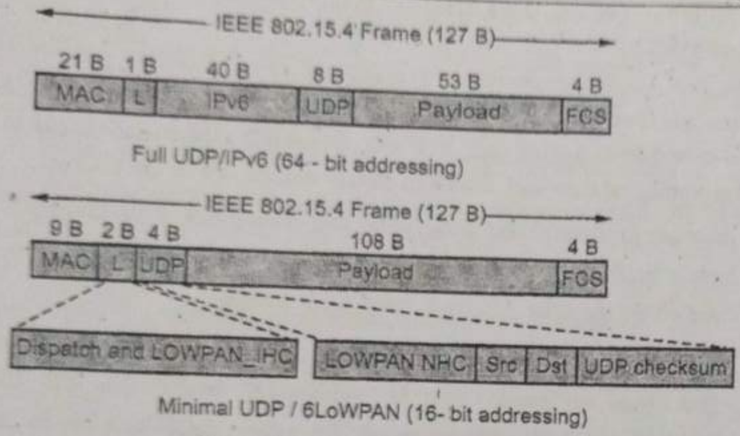


Fig. Q.20.1 Header compression

- Fig. Q.20.1 shows header compression.
- The former exploits the fact that some IPv6 header fields (e.g., IIDs and datagram length) are derivable from the link layer header and an optimistic expectation that several IPv6 header fields will be set to typical values.
- The latter requires 6LoWPAN nodes to share context that allows to substitute address prefixes or complete addresses by a short identifier.
- Subsequently, 6LoWPAN HC has served as the basis for header compression over the 6Lo technologies. 6LoWPAN HC has needed some tailoring to the specific characteristics of each technology, with the exception of IEEE 1901.2, which does not need any changes, since it endorses the IEEE 802.15.4 MAC frame and address formats.

3.5 : Application Layer Protocols

Q.21 What is CoAP ? Explain its key feature.

Ans. : • Constrained Application Protocol (CoAP) is a specialized web transfer protocol for use with constrained nodes and constrained (e.g., low-power, lossy) networks.

- CoAP is designed for simplicity, low overhead and multicast support in resource-constrained environments.
- CoAP is a web protocol that runs over the UDP for IoT. Datagram Transport Layer Security (DTLS) is used to protect CoAP transmission.
- The protocol is designed for Machine-To-Machine (M2M) applications such as smart energy and building automation.
- CoAP provides a request / response interaction model between application endpoints, supports built-in discovery of services and resources and includes key concepts of the Web such as URIs and Internet media types.
- CoAP is designed to easily interface with HTTP for integration with the Web while meeting specialized requirements such as multicast support, very low overhead, and simplicity for constrained environments.
- CoAP is particularly targeted for small low power sensors, switches, valves and similar components that need to be controlled or supervised remotely, through standard Internet networks.
- CoAP is an application layer protocol that is intended for use in resource-constrained internet devices, such as Wireless Sensory Network (WSN) nodes.
- The key features of CoAP are :
  1. CoAP is a RESTful protocol.
  2. Four methods similar to HTTP : Get, Put, Post and Delete.
  3. Four different message types : Confirmable, Non-Confirmable, Acknowledgement and Reset (Nack).
  4. It support synchronous message exchange.
  5. Easy to proxy to and from HTTP.
  6. Constrained web protocol fulfilling M2M requirements.

Q.22 List advantages and disadvantages of CoAP.

Ans. : Advantages :

- It runs over UDP and avoids overhead of TCP.
- It is easy to do HTTP - CoAP translation.
- It is a lightweight application layer protocol designed for constrained devices and constrained networks.

Dec 06, 2023, 19:13

Disadvantages :

- Constraints associated with DTLS.
- No standardized framework for authorization and access control for CoAP exists as of now.
- No explicit support for real-time IoT application at present.

Q.23 Define MQTT ? Explain MQTT characteristics.

Ans. : • Message Queue Telemetry Transport (MQTT) is Open Connectivity for Mobile, M2M and IoT.

- MQTT is designed for high latency, low-bandwidth or unreliable networks. The design principle minimizes the network bandwidth and device resource requirements.
- MQTT is a lightweight broker-based publish / subscribe messaging protocol designed to be open, simple, lightweight and easy to implement.

MQTT characteristics

1. Lightweight message queueing and transport protocol.
2. Asynchronous communication model with messages (events).
3. Low overhead (2 bytes header) for low network bandwidth applications.
4. Publish / Subscribe (PubSub) model.
5. Decoupling of data producer (publisher) and data consumer (subscriber) through topics (message queues).
6. Simple protocol, aimed at low complexity, low power and low footprint implementations.
7. Runs on connection-oriented transport (TCP).
8. MQTT caters for (wireless) network disruptions.

Q.24 Explain MQTT publish/subscribe framework.

Ans. : • Fig. Q.24.1 shows MQTT publish / subscribe framework.

- A producer publishes a message (publication) on a topic (subject). A consumer subscribes (makes a subscription) for messages on a topic (subject).

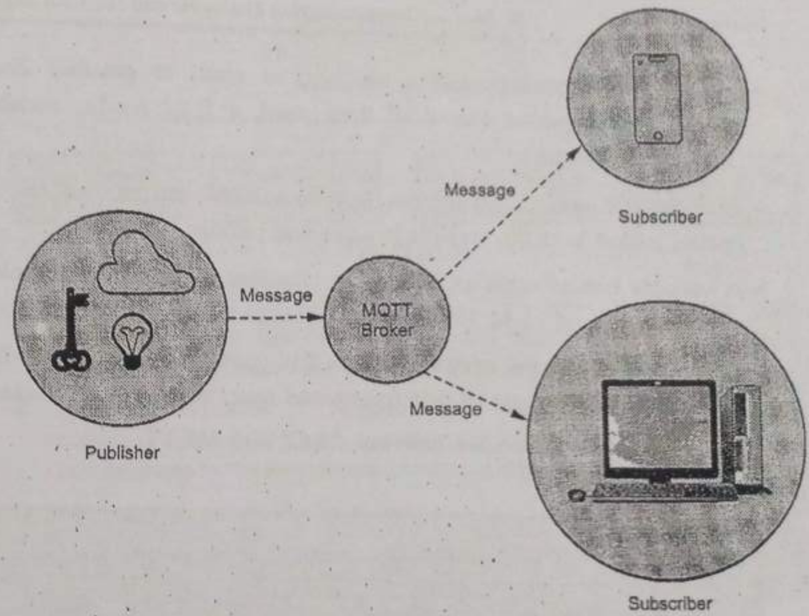


Fig. Q.24.1 MQTT publish/subscribe framework

- A message server (called BROKER) matches publications to subscriptions.
- If none of them match the message is discarded after modifying the topic. If one or more matches the message is delivered to each matching consumer after modifying the topic.
- The MQTT messages are delivered asynchronously ("push") through publish subscribe architecture.
- The MQTT protocol works by exchanging a series of MQTT control packets in a defined way.
- Each control packet has a specific purpose and every bit in the packet is carefully crafted to reduce the data transmitted over the network.
- A MQTT topology has a MQTT server and a MQTT client. MQTT client and server communicate through different control packets. Table below briefly describes each of these control packets.

Dec 06, 2023, 19:13

- MQTT control packet headers are kept as small as possible. Each MQTT control packet consist of three parts, a fixed header, variable header and payload.
- Each MQTT control packet has a 2 byte fixed header. Not all the control packet have the variable headers and payload.
- A variable header contains the packet identifier if used by the control packet. A payload up to 256 MB could be attached in the packets.
- Having a small header overhead makes this protocol appropriate for IoT by lowering the amount of data transmitted over constrained networks.

Q.25 Explain the difference between CoAP and MQTT.

Ans. :

Sr. No.	CoAP	MQTT
1.	CoAP uses UDP protocol.	MQTT uses TCP protocol.
2.	It uses Request/Response messaging.	It uses publish/subscribe messaging.
3.	Communication model is one-to-one.	Communication model is many-to-many.
4.	Advantages : • Lightweight and fast • Low overhead • Support for multicasting	Advantages : • Simple management • Scalability • Robust communication
5.	Weakness : Not as reliable as TCP based MQTT	Weakness : Higher overhead, no multicasting support
6.	Security type is DTLS.	Security type is SSL/TLS.
7.	Effectiveness in LLN is excellent.	Effectiveness in LLN is low.

Q.26 What is AMQP ? Explain architecture of AMQP.

Ans. : • Advanced Message Queuing Protocol (AMQP) is a software layer protocol for message-oriented middleware environments. AMQP is an open protocol for asynchronous message queuing.

- AMQP is an open standard, binary application layer protocol designed for message-oriented middleware i.e., AMQP protocol standardizes messaging using Producers, Brokers and Consumers and messaging increases loose coupling and scalability.
- A protocol to communicate between clients and messaging middleware servers (brokers). The broker is the AMQP Server.
- AMQP supports both publish-subscribe model and point-to-point communication, routing and queuing.
- AMQP divides the brokering task between exchanges and message queues, where the first is a router that accepts incoming messages and decides which queues to route the messages to, and the message queue stores messages and sends them to message consumers.
- AMQP supports username and password authentication as well as SASL authorization. It also supports TLS encryption. Fig. Q.26.1 shows AMQP architecture.

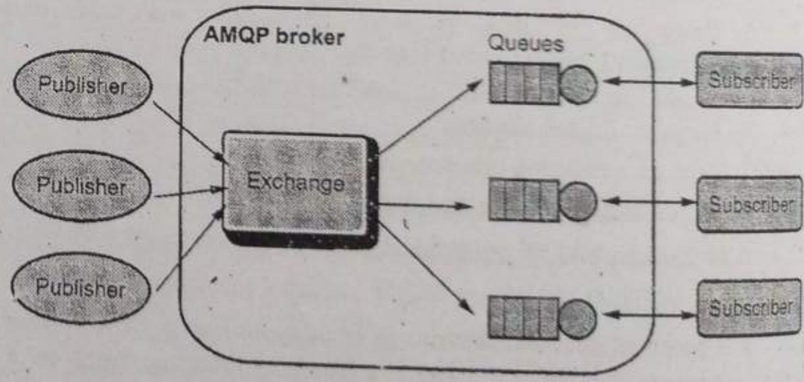


Fig. Q.26.1 AMQP architecture

- **Exchange :** Receives messages from publisher primarily based programs and routes them to message queues.

- **Message Queue :** Stores messages until they may thoroughly process via the eating client software.
- **Binding :** States the connection between the message queue and the change.

3.6 : Wireless Medium Access Issue

Q.27 Which attributes are consider while designing good MAC protocol for WSN ?

Ans. : • To design a good MAC protocol for the wireless sensor networks, the following attributes must be considered :

1. The first attribute is the energy efficiency. We have to define energy efficient protocols in order to prolong the network lifetime.
2. Other important attributes are scalability and adaptability to changes. Changes in network size, node density and topology should be handled rapidly and effectively for a successful adaptation.

- A good MAC protocol should gracefully accommodate such network changes. Other typical important attributes such as latency, throughput and bandwidth utilization may be secondary in sensor networks.

Q.28 What do you mean routing protocols ? Explain basic classification of routing protocol.

Ans. : • The routing protocols define how nodes will communicate with each other and how the information will be disseminated through the network. The basic classification of routing protocols is shown in Fig. Q.28.1.

1. **Node centric :** In node centric protocols the destination node is specified with some numeric identifiers and this is not expected type of communication in Wireless sensor networks. LEACH is a routing protocol that organizes the cluster such that the energy is equally divided in all the sensor nodes in the network.
2. In data centric routing the sink node queries to specific regions to collect data of some specific characteristics so naming scheme based on attributes is necessary to describe the characteristics of data.

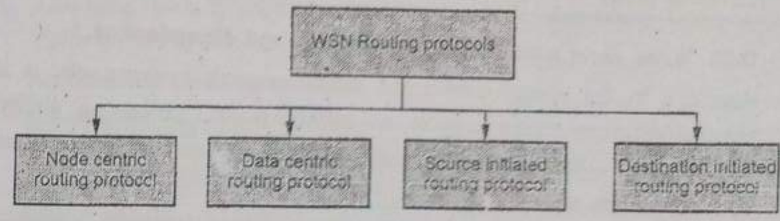


Fig. Q.28.1

3. **Source-initiated :** In these types of protocols the source node advertises when it has data to share and then the route is generated from the source side to the destination. Examples is SPIN.
4. **Destination-initiated :** Protocols are called destination initiated protocols when the path setup generation originates from the destination node. Examples are Directed Diffusion (DD) and LEACH.

Q.29 Discuss sensor node deployment.

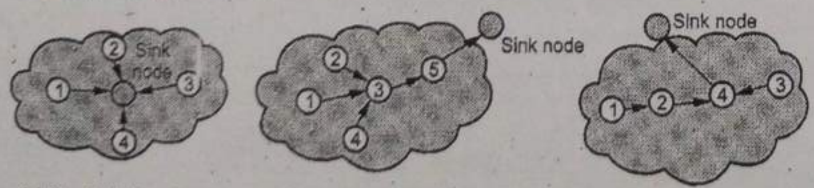
Ans. : • Depending on the application requirements, sensor node deployment can be either random or deterministic.

- Random deployment is normally adopted in large-scale applications, where the sensor nodes are scattered randomly into the network field. For instance, sensors can be dropped from an airplane to a remote and hard-to-access area.
- Random deployment is more applicable in dangerous and inaccessible environments.
- Deterministic deployment is commonly adopted in small to medium-scale applications, where the number and position of the sensor nodes can be predetermined in advance. The sensors can be deployed either manually or by robots. A regular and symmetric deployment/placement pattern is usually adopted in such deployment methods.
- Deterministic deployment is suitable only in controlled and human-friendly environments.

Q.30 Write short note on data aggregation and dissemination.

Ans. : • In the energy-constrained sensor network environments, it is unsuitable in numerous aspects of battery power, processing ability, storage capacity and communication bandwidth, for each node to transmit data to the sink node.

- To avoid the above mentioned problems, data aggregation techniques have been introduced. Data aggregation is the process of integrating multiple copies of information into one copy, which is effective and able to meet user needs in middle sensor nodes.
- The introduction of data aggregation benefits both from saving energy and obtaining accurate information. The energy consumed in transmitting data is much greater than that in processing data in sensor networks.
- Therefore, with the node's local computing and storage capacity, data aggregating operations are made to remove large quantities of redundant information, so as to minimize the amount of transmission and save energy.
- As a result, monitoring the data of the same object requires the collaborative work of multiple sensors which effectively improves the accuracy and the reliability of the information obtained.
- The performance of data aggregation protocol is closely related to the network topology. It is then possible to analyze some data aggregation protocols according to star, tree, and chain network topologies as seen in Fig. Q.30.1.



(a) Star WSN topology (b) Tree WSN topology (c) Chain WSN topology

Fig. Q.30.1

- Data aggregation is any process in which information is gathered and expressed in a summary form for purposes such as statistical analysis.

Aggregated data is normally found in a data warehouse. IoT sensor nodes can also eliminate redundancies in the data received from neighboring nodes before transferring the final data packages.

- Data aggregation technology could save energy and improve information accuracy, while sacrificing performance in other areas. On one hand, in the data transfer process, looking for aggregating nodes, data aggregation operations and waiting for the arrival of other data are likely to increase in the average latency of the network.
- On the other hand, compared to conventional networks, sensor networks have higher data loss rates. Data aggregation could significantly reduce data redundancy but lose more information inadvertently, which reduces the robustness of the sensor network.

END...

Dec 06, 2023, 19:13